



Biology notes By Vasumitra Gajbhiye

CHAPTER 1

- Movement is an action by an organism or part of an organism causing a change of position or place.
 - Respiration is the chemical reactions in cells that break down nutrient molecules and release energy for metabolism.
 - Sensitivity is the ability to detect and respond to changes in the internal or external environment.
 - Reproduction is the processes that make more of the same kind of organism.
 - Excretion is the removal of the waste products of metabolism and substances in excess of requirements.
 - Nutrition is the taking in of materials for energy, growth and development.
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- Organisms can be classified into groups by the features that they share.
 - Species is a group of organisms that can reproduce to produce fertile offspring.
 - Binomial system of naming species is an internationally agreed system in which the scientific name of an organism is made up of two parts showing the genus and species.
 - Classification systems aim to reflect evolutionary relationships.
 - The sequences of bases in DNA are used as a means of classification.
 - Groups of organisms which share a more recent ancestor (are more closely related) have base sequences in DNA that are more similar than those that share only a distant ancestor.

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- There are five kingdoms
 - Animal kingdom
 - Their cells have a nucleus, but no cell walls or chloroplasts.
 - They feed on organic substances made by other living organisms.
 - Plant kingdom
 - Their cells have a nucleus and cell walls made of cellulose and often contain chloroplasts.
 - They feed by photosynthesis.
 - They may have roots, stems and leaves (but some plants do not have these organs).
 - Fungus
 - They are usually multicellular (many-celled), but some such as yeast are unicellular (single-celled).
 - They have nuclei and cell walls, but the walls are not made of cellulose.
 - They do not have chlorophyll.
 - They feed by digesting waste organic material and absorbing it into their cells.
 - Protoctista
 - They are multicellular or unicellular.
 - Their cells have a nucleus and may or may not have a cell wall and chloroplasts.
 - Some feed by photosynthesis and others feed on organic substances made by other organisms.
 - Prokaryote
 - They are usually unicellular (single-celled).
 - They have no nucleus.
 - They have cell walls, not made of cellulose.

- They have no mitochondria.
- They have a circular loop of DNA, which is free in the cytoplasm.
- They often have plasmids.
- Vertebrates are animals that have a backbone:
 - Fish
 - They are vertebrates with scaly skin.
 - They have gills throughout their life.
 - They have fins.
 - Their eggs have no shells and are laid in water.
 - Amphibians
 - They are vertebrates with skin with no scales.
 - Their eggs have no shells and are laid in water.
 - The tadpoles live in water, but adults often live on land.
 - The tadpoles have gills for gas exchange, but adults have lungs.
 - Reptiles
 - They are vertebrates with scaly skin.
 - They lay eggs with soft shells on land.
 - Birds
 - They have feathers (and also sometimes a few scales).
 - They have a beak.
 - Their front two limbs are wings (though not all birds can fly).
 - They lay eggs with hard shells.
 - Mammals
 - They have hair on their skin.
 - Their young develop in a uterus, attached to the mother by a placenta.
 - The females have mammary glands, which produce milk to feed their young.

- They have different kinds of teeth (incisors, canines, premolars and molars).
- They have a pinna (ear flap) on the outside of the body.
- They have sweat glands in the skin.
- They have a diaphragm.
- Arthropods are animals that have jointed legs and no backbone.
 - Insects
 - They are arthropods with three pairs of jointed legs.
 - They have two pairs of wings (one or both may be vestigial).
 - They have one pair of antennae.
 - Their body is divided into a head, thorax and abdomen.
 - They breathe through tubes called tracheae.
 - Crustaceans
 - They are arthropods with more than four pairs of jointed legs.
 - They have two pairs of antennae.
 - Arachnids
 - They are arthropods with four pairs of jointed legs.
 - They have no antennae.
 - Their body is divided into two parts – a cephalothorax and abdomen.
 - Myriapods
 - Their body consists of many similar segments.
 - Each of their body segments has jointed legs.
 - They have one pair of antennae.
- Ferns are plants with leaves called fronds. They do not produce flowers, but instead reproduce by means of spores produced on the underside of the fronds.
 - They are plants with roots, stems and leaves (fronds).
 - They do not produce flowers.

- They reproduce by spores produced on the undersides of their fronds.
 - Flowering plants reproduce by using flowers.
 - They are plants with roots, stems and leaves.
 - They reproduce using flowers and seeds.
 - Their seeds are produced inside an ovary, in the flower
 - Characteristic features of dicots:
 - They have seeds with two cotyledons.
 - They usually have a main root with side roots coming out from it.
 - Their leaves have a network of veins.
 - They have flower parts (e.g. petals) in multiples of four or five.
 - They have vascular bundles in the stem, arranged in a ring
 - Characteristic features of monocots:
 - They have seeds with one cotyledon.
 - Their roots grow out directly from the stem.
 - Their leaves have parallel veins.
 - They have flower parts in multiples of three.
 - They have vascular bundles in the stem, arranged randomly.
 - Viruses are not living. They have a protein coat and genetic material inside without nucleus.
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CHAPTER 2

- **Structure of cell:**
 - **Cytoplasm** is found inside the cell and contains all the other cell structures
 - The large nucleus is surrounded by a **nuclear membrane** to separate it from the cytoplasm
 - The **cell membrane** surrounds the cell

- The **cell wall** is made of cellulose and surrounds the cell membrane in plant cells
- **Chloroplasts** are organelles found in the cytoplasm that are packed with the pigment chlorophyll and so are green in colour
- **Vacuoles** are large vesicles that take up a large part of the interior of plant cell
- **Mitochondria** (singular: mitochondrion) are organelles found throughout the cytoplasm
- **Ribosomes** are tiny structures that can be free within the cytoplasm or attached to a system of membranes within the cell known as Endoplasmic Reticulum
- **Endoplasmic reticulum** studded with ribosomes looks rough under the microscope; this gives rise to its name of Rough Endoplasmic Reticulum (R.E.R.)
- **Vesicles** can also be seen using a higher magnification – these are small circular structures found moving throughout the cytoplasm
- **Function of cell organelles:**
 - Nucleus
 - Contains genetic material in chromosomes which
 - Control how cells grow and work
 - Controls cell division
 - Cytoplasm
 - supports cell structures
 - site of many chemical reactions
 - contains water and many solute
 - Cell membrane
 - holds the cell together
 - controls substances entering and leaving the cell
 - Cell wall
 - gives the cell extra support and defines its shape

- Chloroplast
 - site of photosynthesis, providing food for plants
 - the chlorophyll pigments absorb light energy needed for the reaction to occur
 - Vacuoles
 - contains cell sap
 - used for storage of certain materials
 - also helps support the shape of the cell
 - Mitochondria
 - site of aerobic respiration, providing energy for the cell
 - cells with high rates of metabolism (carrying out many different cell reactions) will have significantly higher numbers of mitochondria than cells with lower numbers of reactions taking place in them
 - Ribosomes
 - site of protein production in protein synthesis
 - Vesicles
 - used to safely transport substances from one part of the cell to another
 - New cells are produced by division of existing cells.
 - Specialised cells have specific functions
 - ciliated cells – movement of mucus in the trachea and bronchi
 - root hair cells – absorption
 - palisade mesophyll cells – photosynthesis
 - neurones – conduction of electrical impulses
 - red blood cells – transport of oxygen
 - sperm and egg cells (gametes) – reproduction
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- Magnification = image size/ actual size

- $1\text{mm} = 1000\mu\text{m}$

CHAPTER 3

- Diffusion is the net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement.
 - The energy for diffusion comes from the kinetic energy of random movement of molecules and ions.
 - Some substances move into and out of cells by diffusion through the cell membrane.
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- The role of water is as a solvent in organisms during digestion, excretion and transport.
 - Water diffuses through partially permeable membranes by osmosis.
 - Water moves into and out of cells by osmosis through the cell membrane.
 - Plants are supported by the pressure of water inside the cells pressing outwards on the cell wall.
 - Osmosis is the net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution), through a partially permeable membrane
 - If a plant cell is placed in a less concentrated solution, water moves into the cell down the water potential gradient by osmosis. pressure on the cell wall increase. turgor pressure increase. cell wall prevents the cell from bursting.
 - If a plant cell is placed in a more concentrated solution, water moves out of the cell down the water potential gradient by osmosis. Pressure on the cell wall decrease. turgor pressure decrease. the cell membrane pulls away from the cell wall. the cell becomes flaccid. The cell is said to be plasmolyzed.
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- Active transport is the movement of particles through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient), using energy from respiration.
 - Protein carriers move molecules or ions across a membrane during active transport.

CHAPTER 4

- Carbohydrates and fats are made up of carbon, hydrogen and oxygen.
- Proteins are made up of carbon, hydrogen, oxygen and nitrogen.
- Starch, glycogen and cellulose are made from glucose.
- Proteins are made from amino acids.
- Fats and oils are made from 3 fatty acids and 1 glycerol.
- Add yellow brown iodine to unknown solution, colour change to blue black if starch is present.
- Add benedict solution to unknown solution, heat in water bath at 85°C, colour change to yellow, green, orange or red if reducing sugar is present.
- Add blue biuret solution to unknown solution, colour change to purple if protein is present.
- Add ethanol to unknown solution, shake well, add this solution to water, emulsification occurs if fat is present.
- Add DCPIP solution to unknown solution, it becomes colourless if vitamin C is present.
- Structure of DNA:
 - Two strands coiled together to form a double helix
 - Each strand contains chemicals called bases
 - Bonds between pairs of bases hold the strands together
 - The bases always pair up in the same way: A with T, and C with G

CHAPTER 5

- A catalyst is a substance that increases the rate of a chemical reaction and is not changed by the reaction.
- Enzymes are proteins that are involved in all metabolic reactions, where they function as biological catalysts.
- Enzymes increase the reaction rate to the necessary reaction rate to sustain life.
- Enzymes and substrate collide. Enzyme and substrate fit together. Substrate binds with Enzyme's active site. Enzyme-substrate complex forms. Because enzymes are

complementary to its substrate. This results in formation of a product.

- Enzymes are specific. Substrate fits into the active site of an enzyme. Therefore, the shape of the enzyme and substrate must be complementary.
- If the temperature is increased, enzymes gain kinetic energy and enzymes move faster, frequency of effective collisions increase. If the temperature increases above the optimum temperature, the enzyme loses its shape and enzyme denatures. The substrate no longer fits into the enzyme active site. And the enzyme can not catalyse the reaction anymore.
- If the temperature is decreased, enzymes lose kinetic energy and move slower, frequency of effective collisions decrease. Therefore, the rate of reaction decrease. Enzymes do not denature in this process.
- Extreme pH can also denature the enzyme.

CHAPTER 6

- Photosynthesis is the process by which plants synthesise carbohydrates from raw materials using energy from light.
- Carbon dioxide + water $\rightarrow$ glucose + oxygen (in the presence of light and chlorophyll)
- Chlorophyll is a green pigment that is found in chloroplasts.
- Chlorophyll transfers energy from light into energy in chemicals, for the synthesis of carbohydrates.
- Uses of glucose:
 - starch as an energy store
 - cellulose to build cell walls
 - glucose used in respiration to provide energy
 - sucrose for transport in the phloem
 - nectar to attract insects for pollination
- Protein contains not only carbon, hydrogen, oxygen, but also nitrogen. Therefore, in order to make amino acids plant need a source of nitrogen, and they get this nitrogen from nitrite ions. They use this nitrate ions to make amino acids and use this amino acids to make protein which is used for growth. If the plant cannot get

enough nitrate ions, it will not be able to synthesise protein effectively. And so will not grow quickly or strongly.

- Chlorophyll is **not** a protein, but it contains nitrogen. It also contains magnesium. Therefore, a plant needs magnesium ions as well as nitrite ions to make chlorophyll. Without these ions, the plant will look yellow and not green, suggesting that there are not plenty of chlorophylls present. If the plant cannot make lots of chlorophyll, it will not be able to photosynthesise well, and will therefore not grow well.
- Limiting factors for photosynthesis are
 - Light intensity.
 - As the light intensity increases, the rate of photosynthesis increases until the plant is photosynthesizing as fast as it can.
 - Carbon dioxide concentration.
 - The more carbon dioxide a plant is given the faster it can photosynthesize. But once the carbon dioxide concentration reaches a certain level, there is no further increase in the rate of photosynthesis.
 - Temperature.
 - As the temperature increases, the rate of photosynthesis is also increased. But after the optimum temperature, the rate of reaction decreases.
 - Stomata.
 - Carbon dioxide diffuses into the leaf through the stomata. If the stomata are closed, photosynthesis cannot take place because the plant lacks one of its raw materials - carbon dioxide. Stomata are often closed during hot weather to prevent water loss. This means that on a really hot or bright day, photosynthesis may slow down.
- Leaves have large surface area and are thin.
 - This large surface area allows large amount of sunlight to fall onto the leaves. The large surface area also increases the rate at which carbon dioxide can diffuse into the leaf from the air.
 - Being thin means that sunlight can pass right through the leaf, allowing many cells inside it to photosynthesise. The thinness also helps carbon dioxide to reach all the cells quite quickly by diffusion.

- Upper epidermis:
 - Cells are tightly packed together to reduce water loss.
 - No chloroplast, so can not photosynthesis.
 - Secret a waxy substance, which forms a thin, transparent, waterproof covering called the cuticle.
 - Transparent to let the light through easily, to reach the palisade mesophyll cells.
- Palisade mesophyll:
 - Tightly packed
 - Column shape.
 - Large number of chloroplasts.
 - Close to the top to get plenty of sunlight.
- Spongy mesophyll:
 - Not many chloroplast.
 - Loosely packed.
 - There are air spaces between them.
 - They allow diffusion of carbon dioxide and oxygen between the air and the cells inside the leaf.
 - This space also allows water vapour to move out of the leaf.
- Lower epidermis:
 - On some leave this tissue makes cuticle But usually it does not because the underside of the leaf does not get as hot as the upper side.
 - Stomata are present. Each stomata is surrounded by a pair of guard cells.
 - Stomata is a small opening, which allow diffusion of carbon dioxide, oxygen and water vapour.
- Guard cells contain chloroplast.

CHAPTER 7

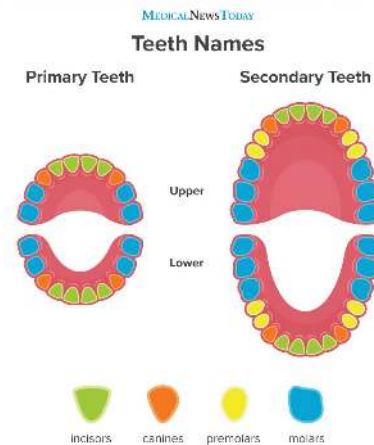
- A balance diet contains correct amount and proportion of carbohydrates, proteins, fats, vitamins, minerals, water, fibre (roughage).
- Principal and dietary sources are:
 - carbohydrates
 - Needed for: energy.
 - Sources: Potatoes, wheat(often made into bread or pasta), rice and maize.
 - fats and oils
 - Needed for: energy, to make cell membrane. Stored under skin in adipose tissue to reduce heat loss from body to air. Provide mechanical protections for organs like kidney by forming a layer around them.
 - Sources: cooking oil, meat, dairy product and oily fish.
 - proteins
 - Needed for: to build new cells for growth to, to make haemoglobin, insulin, antibodies.
 - Sources: meat, fish, eggs, dairy products, peas, beans, nuts and seeds.
 - vitamins, C and D
 - **Vitamin C:**
 - Needed for: to make the stretchy protein collagen found in skin and other tissues. Help tissue in good repair.
 - Source: citrus fruit, such as oranges, lime and raw vegetables.
 - **Vitamin D:**
 - Needed for: helps calcium to be absorbed for making bones and teeth.
 - Sources: oily fish, liver, red meat, egg yolk, sun
 - mineral ions:
 - **Calcium:**
 - Needed for: bones and teeth and blood clotting.
 - Source: Milk and other daily products, bread.

- **Iron:**
 - Needed for: making haemoglobin the red pigment in blood, which carries oxygen.
 - Source. liver, red meat, egg yolk, dark green vegetables.
 - fibre (roughage)
 - Needed for: Absorb and remove, toxins, prevent constipation.
 - Source: all plant food such as fruits and vegetables.
 - water
 - Needed for: An important solvent. Metabolic reaction can only take place in a solution. Plasma is mostly water. Cytoplasm is mostly water. To dissolve enzymes and nutrients in alimentary canal.
 - Source: By drinking fluid and some fruits.
 - Vitamin C deficiency causes scurvy which causes pain in joints and muscles and bleeding from gums and other places. This used to be a common disease for sailors who had no fresh vegetables during long voyages.
 - Vitamin D deficiency causes rickets, in which the bones become soft and deformed; this disease was common in young children in industrial areas, who rarely got out into the sunshine.
 - Calcium deficiency results in brittle bones and teeth; poor blood clotting.
 - Iron deficiency causes anaemia, in which there are not enough red blood cells so the tissues do not get enough oxygen delivered to them.
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- Ingestion – the taking of substances, e.g. food and drink, into the body. Occurs in mouth.
 - digestion – the breakdown of food
 - absorption – the movement of nutrients from the intestines into the blood
 - assimilation – uptake and use of nutrients by cells
 - egestion – the removal of undigested food from the body as faeces.
 - Oesophagus - Food travels down the oesophagus into the stomach.

- Stomach:
 - Contains strong muscular walls. Muscles relax and contract to mix food enzyme and mucus.
 - Goblet cells in the walls secrete mucus.
 - Digest protein.
 - Produce hydrochloric acid. Hydrochloric acid lowers the pH to about 2, which is the optimum pH for protease in stomach. After Stomach, the food moves into duodenum.
 - Small intestine:
 - Duodenum:
 - Mix food with bile and pancreatic juice.
 - Absorption of minerals, and other nutrients begins in the duodenum.
 - Ileum:
 - Absorb digested nutrients into blood.
 - Absorb water.
 - Large intestine:
 - Colon:
 - absorb remaining water
 - Rectum:
 - Store faeces.
 - Salivary gland secretes saliva in mouth, which contains the enzyme amylase, that starts the digestion of starch.
 - Pancreas secretes digestive juices, which helps in digestion of fat.
 - Liver produces bile. It is stored in gallbladder. Bile increases the pH of acidic food from the stomach. And emulsifies fat.
 - More water is absorbed in small intestine than large intestine.
-

- Physical digestion is the breakdown of food into smaller pieces without chemical change to the food molecules.
- Physical digestion increases the surface area of food for the action of enzymes in chemical digestion.

- The tooth is embedded in the gum. It is covered with a hard substance called enamel. Underneath enamel there is a layer of dentine, it is like a hard bone but not as hard as enamel. It has channels in it which contain living cytoplasm. In the middle of the tooth, there are nerves and blood vessels. The blood vessels supply the cytoplasm in the dentine with nutrients and oxygen. The part of the tooth that is embedded in the gum is covered with cement. This has fibres growing out of it.



- Incisors are chisel shaped for biting off pieces of food.
- Canines are very similar to incisors in humans.
- Premolars have wide surfaces for grinding food.
- Molars, like premolars, are used for grinding.
- Muscular walls of the stomach, mix the food, enzyme and mucus together in the process of churning.

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- Chemical digestion is the break down of large insoluble molecules into small soluble molecules.
 - The role of chemical digestion in producing small soluble molecules that can be absorbed.
 - Amylase breaks down starch to maltose. It is secreted by salivary glands in mouth and by pancreas in duodenum
 - Maltase breaks down maltose to glucose. Maltase is secreted by the cells in the lining of the small intestine on the membranes of the epithelium lining of the small

intestine.

- Pepsin breaks down protein in acidic conditions and is secreted by the walls of the stomach in the stomach.
- Trypsin breaks down protein in alkaline conditions and is secreted by the pancreas in the duodenum.
- Lipase breaks down fats and oils to fatty acids and glycerol. It is secreted by the pancreas in the duodenum.
- Function of hydrochloric acid in gastric juice is to kill harmful microorganisms in food and provide an acidic pH for optimum enzyme activity.
- Bile is an alkaline mixture that neutralises the acidic mixture of food and gastric juices entering the duodenum from the stomach, to provide a suitable pH for enzyme action

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- Small intestine is the region where nutrients are absorbed.
 - Most water is absorbed from the small intestine but some is also absorbed from the colon.
 - Villi and microvilli increase the internal surface area of the small intestine and help in faster absorption of dissolved nutrients.
 - Villi are tiny projections on the internal wall of the small intestine. The cell membrane on the surface of each villus is folded to form many tiny microvilli. Villi contains lacteal and blood capillaries.
 - Capillaries in villi absorb glucose, amino acids, fatty acids, glycerol, vitamins, mineral ions and water. These blood capillaries join to form hepatic portal vein, which takes all of these substances to the liver.
 - The fatty acids and glycerol pass into the lacteals. The contents of the lacteals are eventually empties into the blood.

CHAPTER 8

- Function of xylem is to transport water and mineral ions and provide support.
- Function of phloem is to transport sucrose and amino acid.
- Xylem have:

- thick walls with lignin
 - no cell contents
 - cells joined end to end with no cross walls to form a long continuous tube.
-
- The large surface area of root hairs increases the uptake of water and mineral ions.
 - Pathway of water from soil: root hair cells, root cortex cells, xylem, mesophyll cells.
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- Transpiration is the loss of water vapour from leaves.
 - Water evaporates from the surfaces of the mesophyll cells into the air spaces and then diffuses out of the leaves through the stomata as water vapour.
 - Water vapour evaporates from the surface of the mesophyll cell out of the leaf. This reduces the pressure in the vessels. The pressure at the top of the Xylem Vessel is therefore less than the pressure at the bottom in the roots. Due to this pressure difference, water flows upwards. It is called transpiration pull. Because it is caused by loss of water vapour by transpiration.
 - At higher temperature rate of transpiration increases because at higher temperature, molecules have greater kinetic energy, which means the rate of evaporation is higher from the surface of the mass of cells.
 - On a windy day, the air around the leaf – which contains a lot of water vapour that has just diffused out of the leaf – is quickly moved away. This means that there is always a diffusion gradient for the water vapour, because there is less of it outside the leaf than in the air spaces inside the leaf. So, water vapour diffuses out of the leaf faster on a windy day than on a day when the air is still.
 - As humidity increase, rate of transpiration decrease because there is a smaller diffusion gradient for the water between the air and inside leaf.
 - Cells lose water by transpiration down the water potential gradient by osmosis. The pressure of water on the cell wall decreases. The turgor pressure decreases. The leaf becomes soft and floppy. This is called wilting.
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- Translocation is the movement of sucrose and amino acids in phloem from sources to sinks.
 - Sinks are the parts of plants that use or store sucrose or amino acids.

- Sources is the parts of plants that release sucrose or amino acids.
- During summers when sunlight is available plant photosynthesis, in this case, the leaves are the source and the roots are the sink as they are the storage of starch. During winters, the plant cannot photosynthesize. In this case, the roots are the source and leaves are the sink because root contains stored starch, which can be converted to glucose and used by the leaves.

CHAPTER 9

- The circulatory system is a system of blood vessels with a pump and valves to ensure one-way flow of blood.
- In single circulation of fish. The blood passes through the heart once for one complete circuit. It moves from heart to the gills and then to the rest of the body.
- Mammals have double circulatory system. One pulmonary loop and one systematic loop happens in mammals. Blood passes through the heart twice for one complete circuit.
- Advantage of double circulation
 - high(er), blood pressure / flow rate (than single circulation) ;
 - allows different blood pressure in each loop ;
 - prevent mixing of oxygenated and deoxygenated blood ;
 - allows animals to have high metabolic rates ;
 - allows animals to be, large / tall ;

OR

1. oxygenated and deoxygenated blood, are kept separate / do not mix / separated by septum ;
2. ensures efficient supply of oxygen (to, body / AW) ;
3. ensures efficient supply of (named) nutrients (to, body / AW) ;
4. low(er) pressure in, pulmonary, artery / circuit / AW ;
5. to prevents damage to (capillaries in the) lungs ;
6. allows more time for gas exchange ;

7. allows high(er) pressure (in body) ;
 8. to ensure efficient, blood supply to (rest of) body ;
 9. to allow filtration in kidneys (for excretion) ;
 10. to allow / maintain, a high, metabolic rate / rate of respiration ;
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- Blood is pumped away from the heart in arteries and returns to the heart in veins.
- The activity of the heart may be monitored by: ECG, pulse rate and listening to sounds of valves closing.
- Coronary heart disease is the blockage of coronary arteries. Causes: diet, lack of exercise, stress, smoking, genetic predisposition, age and sex.
- Balanced diet and exercise helps to reduce the risk of coronary heart disease. They reduce stress, lower blood pressure, make heart muscles stronger.
- Atria receives blood from either the lungs or the body and supply it to the ventricles. The ventricles pump blood out of the heart and all around the body. So pressure applied by the ventricles must be high. To help them do this, the ventricles have much thicker, more muscular walls than the atria.
- The right ventricle pumps blood to the lungs, which are very close to the heart. The left ventricle, however, pumps blood to the rest of the body. The left ventricle has an especially thick wall of muscle to enable it to do this.
- Septum separates oxygenated and deoxygenated blood in the heart.

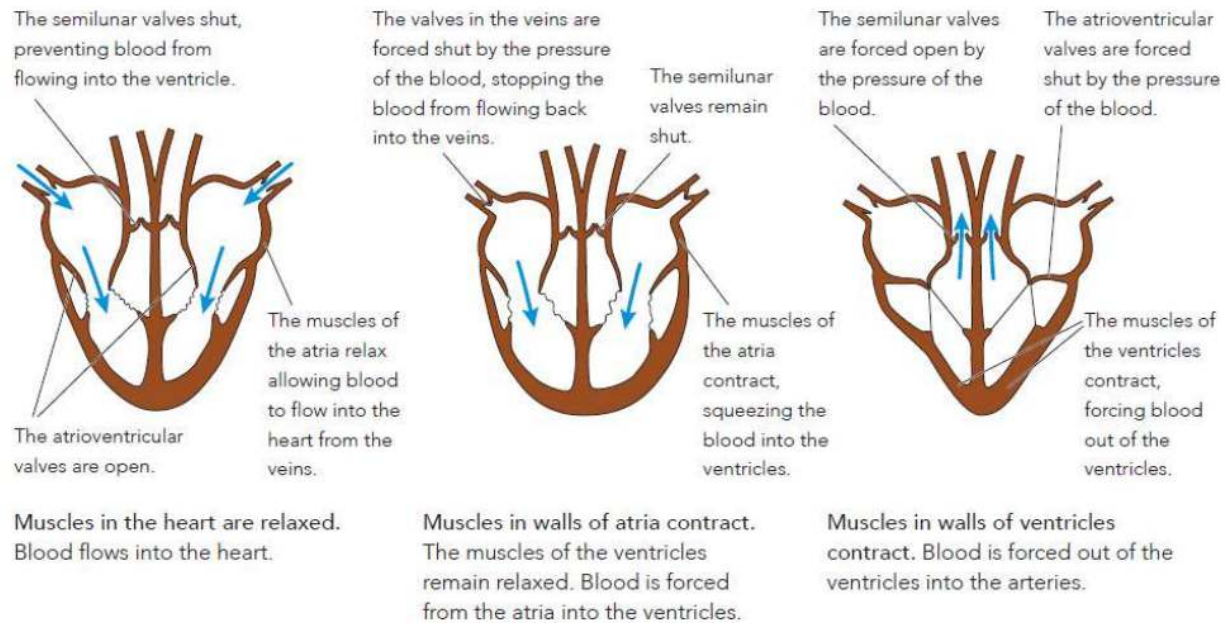
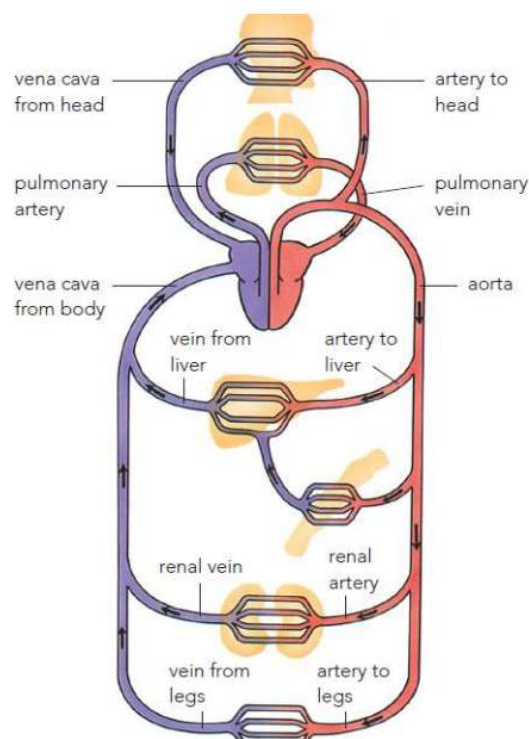


Figure 9.10: How the heart beats.

- As physical activity increase heart rate increase because muscles contract more. Muscles require more energy. More aerobic respiration takes place. More oxygen is required. More blood is required to be transported to muscles for more oxygen. Heart beats faster to supply more oxygen to the muscles.
- Arteries:
 - Have thick walls
 - Diameter of lumen is small
 - No valves are present
- Veins:
 - Have relative thin walls
 - Diameter of lumen is large
 - Valves are present to prevent back flow of blood.
- Blood pressure in arteries is high because it has just been forced out of the heart by the contraction of muscles of the ventricles. Arteries need very strong walls to withstand this high pressure.

- Blood in arteries doesn't flow smoothly, it pulses through. When ventricles contract pressure in arteries is high and when ventricles relax pressure in arteries is reduced. Arteries have elastic tissues in their walls which can stretch and recoil with the force of the blood making the blood flow smoother.
- Capillaries are very small and penetrate to every part of the body so that they can reach every cell.
- Function of the capillaries is to take nutrients, oxygen and other materials to all the cells in the body, and to take away their waste materials. To do this, their walls must be very thin so that substances can get in and out of the capillaries easily.
- Veins function is to return blood to the heart.
 - Walls of veins (muscle and elastic tissue) is thin because there is no need for strong walls as most of the blood pressure has been lost.
 - They have a wide lumen because it offers less resistance to blood flow.
 - Contain valves, they prevent backflow of blood.



- Components of blood are red blood cells, white blood cells, platelets and plasma.

- Function of components of blood:
 - Red blood cells in transporting oxygen. Haemoglobin is present in RBC and it combines with oxygen transports it.
 - White blood cells in phagocytosis and antibody production
 - Platelets in clotting
 - Plasma in the transport of blood cells, ions, nutrients, urea, hormones and carbon dioxide
- The roles of blood clotting are preventing blood loss and prevent entry of pathogens.
- Lymphocytes have one big nucleus which covers almost the entire cell. Phagocytes have nucleus that divides into small connected balls.
- Function of lymphocytes is antibody production. Function of phagocytes is engulfing pathogens by phagocytosis.
- Blood clotting:
 - platelets get attracted towards the damaged blood vessel.
 - enzyme thrombin converts soluble fibrinogen into insoluble fibrin. This forms a fibre mesh like structure.
 - red blood cells and platelets get trapped in the mesh and stop the blood loss.

CHAPTER 10

- Pathogen is a disease-causing organism.
- Transmissible disease is a disease in which the pathogen can be passed from one host to another.
- A pathogen is transmitted:
 - by direct contact, through blood and other body fluids.
 - by indirectly contact, from contaminated surfaces, food, animals and air.
- Body's defence system include: skin, hairs in the nose, mucus, stomach acid and white blood cells.
- To prevent spreading of diseases use:

- a clean water supply - water from dirty rivers or lakes can contain many different pathogens.
- hygienic food preparation - bacteria can grow on the food too, houseflies have harmful bacteria on their feet and they can land on the food to contaminate it.
- good personal hygiene - we should clean our body regularly because oil, dirt and sweat can build up on the skin and this will provide breeding ground for harmful bacteria.
- waste disposal - harmful bacteria can grow on organic waste so it must be disposed as soon as possible.
- sewage treatment - treating the water before drinking removes dirt and microorganisms from it.
- Active immunity is defence against a pathogen by antibody production in the body.
- Each pathogen has its own antigens, which have specific shapes.
- Antibodies are proteins that bind to antigens leading to direct destruction of pathogens or marking of pathogens for destruction by phagocytes.
- Specific antibodies have complementary shapes which fit specific antigens.
- Active immunity is gained after an infection by a pathogen or by vaccination.
- Process of vaccination:
 - weakened pathogens or their antigens are injected into the body
 - the antigens stimulate an immune response by lymphocytes which produce antibodies
 - memory cells are produced that give long-term immunity.
- Vaccination provide active immunity to individuals and prevents pathogens from breeding in them. If 90% of the people are vaccinated to a disease, the pathogen will have very few places to live and grow therefore the spread of disease will decrease. Non-vaccinated individuals will also get protection as a result of herd immunity.
- Passive immunity is a short-term defence against a pathogen by antibodies acquired from another individual, e.g. across the placenta and in breast milk.

- A young baby's immune system is not well developed, and so the mother's antibodies can protect it against any diseases to which she is immune, for the first few months of its life.
- Memory cells are not produced in passive immunity.
- Cholera is a disease caused by a bacterium which is transmitted in contaminated water.
- The cholera bacterium produces a toxin that causes secretion of chloride ions into the small intestine, causing osmotic movement of water into the gut, causing diarrhoea, dehydration and loss of ions from the blood.

CHAPTER 11

- Features of gas exchange surface area in humans:
 - large surface area
 - thin surface
 - good blood supply
 - good ventilation with air

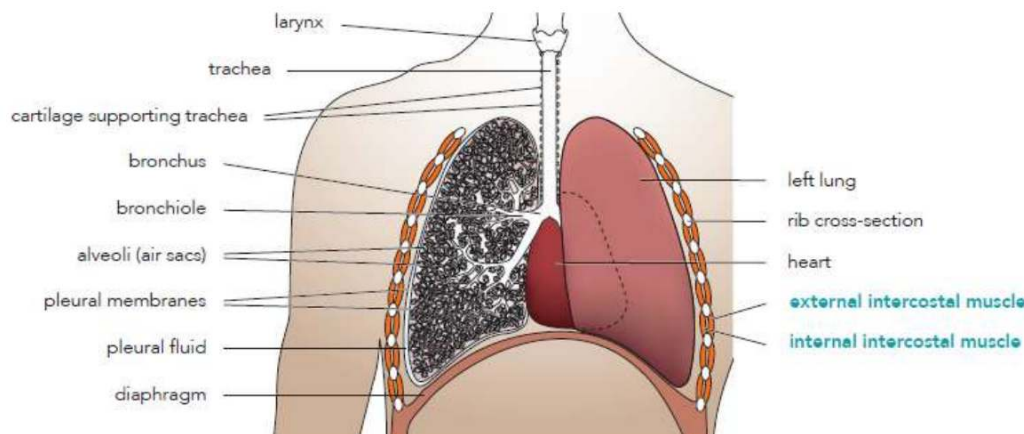


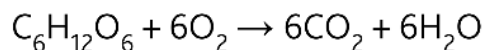
Figure 11.3: The human gas exchange system.

Component	Percentage in inspired air	Percentage in expired air
oxygen	21	16
carbon dioxide	0.04	4
water vapour	variable	usually very high

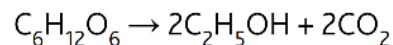
Expired air contains less oxygen because oxygen is used up by the cells in the body. Expired air contains more carbon dioxide because it is a waste product of respiration. It contains more water vapour because the water from the lining of the windpipe evaporate and is expired.

- Function of cartilage is
 - keep the airway/ trachea open
 - prevent collapse
 - protects (named) airways ;
 - allows (free flow of) air into (the lungs) ;
 - allows flexibility / can breathe even when, bent / swallowing / AW ;
 - reduces resistance to movement of air ;
 - sound production in larynx ;
 - forms incomplete rings around, trachea / bronchi ;
- During inspiration: rib cage moves upwards and outwards, diaphragm contract, external intercostal muscles contract, internal intercostal muscles relax, volume of thorax increase, pressure in thorax decrease, air move into the lungs to equalize the pressure.
- During expiration: rib cage moves downwards and inwards, diaphragm relax, external intercostal muscles relax, internal intercostal muscles contract, volume of thorax decrease, pressure in thorax increase, air move out the lungs to equalize the pressure.
- As physical activity increase breathing rate increase because muscles contract more. Muscles require more energy. More aerobic respiration takes place. More carbon dioxide is produced, pH of blood decrease because carbon dioxide is acidic. This is detected by the brain, brain send impulses to breathing muscles like heart and diaphragm to contract more frequently and more hardly. This leads to an increased rate and greater depth of breathing.

- Goblet cells produce mucus. Mucus is a sticky substance which traps pathogens and dust particles and prevent them from entering the lungs. Ciliated cells beat and move the mucus up towards the throat where it can be sneezed out of the body.
- Uses of energy in living organism:
 - muscle contraction
 - protein synthesis
 - cell division
 - active transport
 - growth
 - the passage of nerve impulses
 - the maintenance of a constant body temperature
- Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules to release energy. glucose + oxygen → carbon dioxide + water



- Anaerobic respiration is the chemical reactions in cells that break down nutrient molecules to release energy without using oxygen.
- Anaerobic respiration releases much less energy per glucose molecule than aerobic
- Word equation for anaerobic respiration in yeast: glucose → alcohol(ethanol) + carbon dioxide



- Word equation for anaerobic respiration in muscles during vigorous exercise as: glucose → lactic acid
- Lactic acid builds up in muscles and blood during vigorous exercise causing an oxygen debt.
- To remove oxygen debt after exercise:

- fast heart rate continues to transport lactic acid in the blood from the muscles to liver.
 - deeper and faster breathing continues to supply oxygen for the aerobic respiration of lactic acid.
 - aerobic respiration of lactic acid in the liver.
-

CHAPTER 12

- Electrical impulses travel along neurones.
- Mammalian nervous system consist of :
 - The central nervous system (CNS) consisting of the brain and the spinal cord
 - The peripheral nervous system (PNS) consisting of the nerves outside of the brain and spinal cord.
- Role of the nervous system is coordination and regulation of body functions.
- Reflex arc: receptors detect stimulus and generate impulse, it travels to sensory neurone, then relay neurone, then to motor neuron, then to the effector.
- A reflex action is a means of automatically and rapidly integrating and coordinating stimuli with the responses of effectors (muscles and glands).
- A synapse is a junction between two neurones.
- Structure of synapse include:
 - presence of vesicles containing neurotransmitter molecules
 - Synaptic gap/cleft
 - receptor protein
- Events at a synapse:
 - An impulse stimulates the release of neurotransmitter molecules from vesicles into the synaptic gap.
 - The neurotransmitter molecules diffuse across the gap
 - Neurotransmitter molecules bind with receptor proteins on the next neurone
 - An impulse is then stimulated in the next neurone.

- Synapses ensure that impulses travel in one direction only.
-
- Sense organs are groups of receptor cells responding to specific stimuli: light, sound, touch, temperature and chemicals.
 - Function of each part of eye:
 - cornea – refracts light
 - iris – controls how much light enters the pupil
 - lens – focuses light on to the retina
 - retina – contains light receptors, some sensitive to light of different colours
 - optic nerve – carries impulses to the brain
 - As light intensity increase, diameter of pupil decrease so that less light enters the eye and prevent damage. The antagonistic muscles, circular muscles contract and radial muscles relax.
 - As light intensity decrease, diameter of pupil increase so that more light enters the eye and prevent damage. The antagonistic muscles, circular muscles relax and radial muscles contract.
 - Distant object: Ciliary muscles relax, suspensory ligament pulled tight (tension on suspensory ligament increase). the lens becomes thin/ less convex.
 - Near object: Ciliary muscles contract, suspensory ligament slackened (tension on suspensory ligament decrease). the lens becomes thick/ more convex.
 - There are three different kinds of cone cells (red, green and blue) and two different types of rod cells.
 - Fovea is almost entirely cones, packed tightly together. Rods are found further out on the retina and are less tightly packed.
 - Rod cells are more sensitive at night.
-
- A hormone is a chemical substance, produced by a gland and carried by the blood, which alters the activity of one or more specific target organs.
 - Adrenal gland secrete adrenaline. Pancreas secrete insulin and glucagon. Testes secrete testosterone. Ovaries secrete oestrogen.

- Effect of adrenaline:
 - increased breathing rate
 - increased heart rate
 - increased pupil diameter
 - increase in blood glucose concentration
 - Speed of action of nervous system is faster than hormonal control system. Duration of effect of hormonal control system is longer than the nervous system.
-

- Homeostasis is the maintenance of a constant internal environment.
- Insulin decreases blood glucose concentration.
- In negative feedback:
 - There is a set point – a normal level that the system tries to maintain.
 - There is a 'measuring device' that keeps track of whether the level is within the range of the set point.
 - If the level goes outside the set point, this triggers events to happen that bring the level back into line again.
- If blood glucose concentration is too high, it is detected by the pancreas, pancreas release insulin, insulin stimulate liver cells to absorb glucose from blood across the partially permeable membrane. Liver cells convert glucose to glycogen and store glycogen in liver cells, this results in reduction of blood glucose concentration until it reaches the set, normal point.
- If blood glucose concentration is too low, it is detected by the pancreas, pancreas release glucagon, glucagon stimulate liver cells to convert store glycogen into glucose, and release this glucose into the blood. This results in increase of blood glucose concentration until it reaches the set, normal point.
- Treatment of type 1 diabetes:
 - insulin ;
 - by injection / insulin pump ;
 - regular blood glucose tests ;

- regular meals and controlled diet ;
 - If the internal temperature is above normal, it is detected by the brain. Vasodilation of arterioles occur, more blood flows to the skin, so more heat is lost from the blood, decreasing the internal temperature. Sweating occurs which gives a cooling effect and decreases the internal body temperature.
 - If the internal temperature is below normal, it is detected by the brain. Vasoconstriction of arterioles occur, less blood flows to the skin, so less heat is lost from the blood, increasing the internal temperature. Adipose tissue in skin act as insulation and reduce heat loss. Shivering produce heat and help to keep the body warm.
-
- Gravitropism is a response in which parts of a plant grow towards or away from gravity.
 - Phototropism is a response in which parts of a plant grow towards or away from the direction of the light source.
 - Phototropism and gravitropism of a shoot are examples of the chemical control of plant growth.
 - How auxin work:
 - auxin is made in the shoot tip
 - auxin diffuses through the plant from the shoot tip
 - auxin is unequally distributed in response to light and gravity
 - auxin stimulates cell elongation
-

CHAPTER 13

- Carbon dioxide is excreted through the lungs. The kidneys excrete urea and excess water and ions.
- The function of glomerulus is ultrafiltration from the blood of water, glucose, urea and ions.
- The function of nephron is the reabsorption of all the glucose, some of the ions and most of the water back into the blood.
- Urine contains urea, excess water and excess ions.

- Role of liver is the assimilation of amino acids by converting them to proteins.
- Urea is formed in the liver from excess amino acids.
- Deamination is the removal of the nitrogen-containing part of amino acids to form urea.
- Urea is toxic, it can lower the pH of the body and denature the enzyme. Denaturing of enzyme might lead to death of the individual therefore excretion of urea is important.

DRUGS

- Drug is any substance taken into the body that modifies or affects chemical reactions in the body.
- Antibiotics are used to treat bacterial infections.
- Some bacteria are resistant to antibiotics which reduces the effectiveness of antibiotics.
- Antibiotics kill bacteria but do not affect viruses.
- If antibiotics are taken in only when essential it can limit the development of resistant bacteria.

CHAPTER 14 & CHAPTER 15

- Asexual reproduction is a process resulting in the production of genetically identical offspring from one parent.
- **Advantages** of asexual reproduction:
 - wild species that are adapted to the environment can survive.
 - isolated wild species can reproduce because only single parent is required.
 - crop plants are genetically identical therefore they have the same yield and characteristics.
- **Disadvantage** of asexual reproduction:
 - less variation
 - lack of diversity in wild plants
 - risk of overpopulation in wild plants
 - competition for resources as all individuals are close together in wild plants

- require less energy.
 - quick
 - no gametes
 - favourable characteristics of parent passed on
 - dense colonies outcompete other species
 - no pollinators required
 - reliable
 - no harmful variation
 - wild plants and crops are less likely to survive to change environment
 - no adaptive features in wild plants
 - all individuals are susceptible to the same, diseases / pests in wild plants
 - higher risk of extinction in wild plants
 - no evolution in wild plants
 - genetically identical so higher risk of inheriting genetic diseases in both wild and crop plants
-
- Sexual reproduction is a process involving the fusion of the nuclei of two gametes to form a zygote and the production of offspring that are genetically different from each other.
 - Fertilisation is the fusion of the nuclei of gametes.
 - Nuclei of gametes are haploid and the nucleus of a zygote is diploid.
 - Advantages of sexual reproduction
 - maintain/ increase, population in wild plants
 - allow variation in wild plants
 - allow adaptation to, new/ changed environment(s) in wild plants
 - allow natural selection/ evolution/ formation of new species in wild plants
 - random fusion of gametes in wild plants
 - ability to express recessive traits in wild plants
 - Disadvantages of sexual reproduction
 - Slow
 - Lots of energy is needed
 - Need 2 parents

- allows to use selective breeding in crop plants

- Function of parts of flower:

- Sepals: Protect the flower while it is a bud.
- Petals: Brightly coloured and attract insects to flowers.
- Stamen: It is male part of flower, consists of long filament and anther at the top which contain pollen grain, which contain male gametes.
- Carpel: It is the female part of flower. It contains ovary, which contains ovules, which contains female gamete. At top is style with stigma at the top. function of stigma is to catch pollen grains.

Insect-pollinated flower	Wind-pollinated flower
large, conspicuous petals, often with guide-lines	small, inconspicuous petals, or no petals at all
often strongly scented	no scent
often have nectaries at the base of petals	no nectaries
anthers inside flower, where insect has to brush past them to reach nectar	anthers dangle outside the flower, where they catch the wind
stigma inside flower, where insect has to brush past it to reach nectar	stigmas large and feathery and dangle outside the flower, where pollen in the air may land on it
sticky or spiky pollen grains, which stick to insects	smooth, light pollen, which can be blown in the wind
quite large quantities of pollen made, because some will be eaten or will be delivered to the wrong kind of flower	very large quantities of pollen made, because most will be blown away and lost

- Pollination is the transfer of pollen grains from an anther to a stigma.
- Self-pollination is the transfer of pollen grains from the anther of a flower to the stigma of the same flower or a different flower on the same plant.
- Cross-pollination is the transfer of pollen grains from the anther of a flower to the stigma of a flower on a different plant of the same species.

- Self-pollination doesn't produce lost of variation. Less capacity to respond to changes in the environment. They don't rely on pollinators.
 - Cross-pollination produce variation. More capacity to respond to changes in the environment. They rely on pollinators.
 - Fertilisation occurs when a pollen nucleus fuses with a nucleus in an ovule.
 - Insect pollinated plant have, large, brightly coloured petals, strong scent, nectar to attract insects towards them. The anther and stigma are inside the flower so that the insect brush past them and some pollen stick onto them. The insect then goes to another flower, looking for more nectar. Pollen grains are sticky and spiky so, some of the pollen it picked up at the first flower sticks onto the stigma of the second flower when the insect brushes past it. The stigma is sticky, and many pollen grains get stuck on it.
 - Wind-pollinated flowers produce much more pollen than insect pollinated flowers. This is because a lot of the pollen doesn't land on another flower, so huge amounts are wasted. Usually no petals because there is no need to attract insects. Their anthers and stigmas dangle outside the flower, to catch the wind. The filaments of the anthers are very flexible, so they can swing in the wind and release their pollen. The feathery stigmas have a large surface area, increasing the chances of catching pollen.
- | | |
|---|---|
| <ul style="list-style-type: none"> • Events which occur between pollination and fertilisation. <ul style="list-style-type: none"> ◦ pollen lands on stigma ◦ pollen (grain) germinates/ pollen (grain) grows pollen tube ◦ pollen tube grows down the style ◦ reaches the ovule ◦ (tip of) pollen tube breaks open ◦ male gamete travels down the pollen tube ◦ male gamete enter ovule | <ul style="list-style-type: none"> • Events which occur between fertilisation and production of seed. <ul style="list-style-type: none"> ◦ zygote (is formed) ◦ divides by mitosis ◦ to form embryo ◦ formation of radicle and plumule ◦ formation of, cotyledons/ seed leaf/ food reserve ◦ formation of teste/ seed coat |
|---|---|

- (male gamete) fuse with female gamete
- zygote forms.
- ref to endosperm
- seed formed from ovule

-
- Testes produce sperm. Scrotum protects testes, as well as providing a sort of climate control system. Sperm duct carries sperm away from testis, it then join up with the urethra just below the bladder. Urethra can carry both urine and sperm at different time. Prostate gland makes semen, a fluid in which sperm can swim in. Function of penis is to enter the vagina and release sperm.
 - Ovaries produce eggs or egg cells. Oviducts lead the egg to uterus. Uterus nourish the developing fetus prior to birth. Cervix allows fluids to pass between uterus and vagina. Vagina is the site where sperms are placed after ejaculation.
 - Fertilisation is the fusion of the nuclei from a male gamete (sperm) and a female gamete (egg cell).
 - Adaptation of sperm:
 - flagellum to swim
 - lots of mitochondria to provide energy for swimming by respiration.
 - enzymes in the acrosome to digest its way through the jelly coat.
 - Adaptation of egg cell:
 - energy store to provide energy and nutrients for the growth and development of zygote after fertilisation.
 - jelly coat hardens after the entry of one sperm to prevent entry of multiple sperms.
 - Sperm cells are lot smaller than egg cells. Sperm cells are streamlined and egg cells are round. Sperm cells can move by themselves but egg cells cannot. There are millions of sperms deposited at the vagina but there are maximum of 2 egg cells in female reproductive system.
 - In early development, the zygote forms an embryo which is a ball of cells that implants into the lining of the uterus.

- Placenta connects the embryo to wall of uterus. Placenta is where substances are exchanged between the mother's blood and the embryo's blood.
- **Role of placenta in humans:**
 - gas exchange
 - transfer of dissolved nutrients, from maternal (circulation)/ to fetal.
 - transfer of excretory products, from fetal/ to maternal;
 - by diffusion
 - produces/ secretes, (named) hormones;
 - passive immunity/ antibodies, from maternal/ to fetal
 - prevents/ limits, mixing of blood
 - regulating blood pressure
 - maternal/ fetal **attachment** point
 - red to counter current flow/ maintains concentration gradient
- **Role of amniotic fluid:**
 - protection from (mechanical) shock (of fetus)
 - maintains (constant) temperature (of fetus)
 - allows movement (of fetus)
 - prevents dehydration
- **Amniotic fluid and the amniotic sac:**
 - maintains temperature ;
 - mechanical shock protection ;
 - allows movement (of the fetus) ;
 - provides support (of the fetus) ;
 - provides a sterile environment / prevents infections ;
 - (movement) allows for development of bones and muscles ;
 - ref. to swallowing (of fluid) ;

- lubrication / AW ;
 - Some pathogens and toxins can pass across the placenta and affect the fetus.
-
- Testosterone cause the secondary sexual characters to develop in boys, like growth of facial hair and pubic hair, broadening shoulders and general muscular development and a deepening voice.
 - Oestrogen cause the secondary sexual characters to develop in girls, like increase in size of breast, growth of pubic hairs, hips become wider.
 - Site of production of oestrogen and progesterone in the menstrual cycle is the ovaries. After ovulation site of progesterone production is corpus luteum. During pregnancy site of progesterone production is the placenta.
 - Changes to the lining of uterus during menstrual cycle:
 - loss / thinning, of lining (of uterus), at the beginning of the cycle / during first week / between day 1–7 ;
 - regrowth / thickening of, lining (of uterus), during second week / after loss (of lining) / before ovulation ;
 - thickness of lining remains constant, in the last two weeks / at the end of the cycle / after ovulation / if no fertilisation ;
 - **The role of FSH.**
 - FSH stimulates, development / growth / maturation, of follicles (in ovary) ;
 - FSH stimulates, development / growth / maturation, of, egg (cell) / ova / ovum / (female) gamete(s) ;
 - FSH stimulates, production / release / secretion, of oestrogen (from ovary) ;
 - **The role of LH.**
 - (FSH and) LH (surge) stimulates, ovulation / release of egg from ovary
 - LH stimulates (corpus luteum / yellow body / empty follicle), secretion / production / release, of progesterone
 - decrease in LH decreases secretion of progesterone (by corpus luteum) ;
 - **The role of oestrogen.**

- Stimulates the uterus to develop a lining (to replace the lining lost during menstruation)
 - Post-ovulation, inhibits FSH and LH production in the pituitary gland.
 - **The role of progesterone.**
 - maintains and thickens lining of the uterus
 - inhibits FSH and LH production
 - if fertilisation doesn't occur, levels drop and menstruation occurs.
-
- Sexually transmitted infection (STI) is an infection that is transmitted through sexual contact.
 - Human immunodeficiency virus (HIV) is a pathogen that causes an STI.
 - HIV infection may lead to AIDS.
 - Methods of transmission of HIV:
 - sexual contact
 - sharing contaminated needles
 - sharing contaminated syringes
 - blood transfusion
 - breast milk
 - tissue/ organ transplant
 - HIV is **not** spread through saliva
 - Methods to stop spread of STIs:
 - never have more than one sexual partner
 - use condoms
-

CHAPTER 16

- Chromosomes are made of DNA, which contains genetic information in the form of genes.
- A gene is a length of DNA that codes for a protein.

- An allele is an alternative form of a gene.
 - Females have XX chromosomes and males have XY chromosomes.
 - The sequence of bases in a gene determines the sequence of amino acids used to make a specific protein.
 - Different sequences of amino acids give different shapes to protein molecules.
 - DNA controls cell function by controlling the production of proteins, including enzymes, membrane carriers and receptors for neurotransmitters.
 - How protein is made:
 - the gene coding for the protein remains in the nucleus
 - messenger RNA (mRNA) is a copy of a gene
 - mRNA molecules are made in the nucleus and move to the cytoplasm
 - the mRNA passes through ribosomes
 - the ribosome assembles amino acids into protein molecules
 - the specific sequence of amino acids is determined by the sequence of bases in the mRNA
 - Most body cells in an organism contain the same genes, but many genes in a particular cell are not expressed because the cell only makes the specific proteins it needs. For instance, the cells in salivary gland don't need to make protease but they have the gene for it which is not expressed because the cell doesn't need it.
 - A haploid nucleus is a nucleus containing a single set of chromosomes.
 - A diploid nucleus is a nucleus containing two sets of chromosomes.
 - In a diploid cell, there is a pair of each type of chromosome and in a human diploid cell there are 23 pairs.
-
- Mitosis is nuclear division giving rise to genetically identical cells.
 - Role of mitosis is in growth, repair of damaged tissues, replacement of cells and asexual reproduction.
 - The exact replication of chromosomes occurs before mitosis.

- During mitosis, the copies of chromosomes separate, maintaining the chromosome number in each daughter cell.
 - Stem cells are unspecialised cells that divide by mitosis to produce daughter cells that can become specialised for specific functions.
-
- Meiosis is involved in the production of gametes.
 - Meiosis is a reduction division in which the chromosome number is halved from diploid to haploid resulting in genetically different cells.
-
- Inheritance is the transmission of genetic information from generation to generation.
 - Genotype is the genetic make-up of an organism.
 - Phenotype is the observable features of an organism.
 - Homozygous is having two identical alleles of a particular gene.
 - Two identical homozygous individuals that breed together will be pure-breeding.
 - Heterozygous is having two different alleles of a particular gene.
 - Heterozygous individual will not be pure-breeding.
 - A dominant allele is an allele that is expressed if it is present in the genotype.
 - A recessive allele is an allele that is only expressed when there is no dominant allele of the gene present in the genotype.
 - Codominance is a situation in which both alleles in heterozygous organisms contribute to the phenotype.
 - A sex-linked characteristic is a feature in which the gene responsible is located on a sex chromosome and that this makes the characteristic more common in one sex (usually male) than in the other.
 - Red-green colour blindness is an example of sex linkage.
-

CHAPTER 18

- Variation is differences between individuals of the same species.
- Continuous variation results in a range of phenotypes between two extremes; examples include body length and body mass.

- Discontinuous variation results in a limited number of phenotypes with no intermediates; examples include ABO blood groups, seed shape in peas and seed colour in peas
- Discontinuous variation is usually caused by genes only and continuous variation is caused by both genes and the environment.
- Mutation is a genetic change.
- Mutation is the way in which new alleles are formed.
- Ionising radiation and some chemicals increase the rate of mutation.
- Gene mutation is a random change in the base sequence of DNA.
- Mutation, meiosis, random mating and random fertilisation are sources of genetic variation in populations.

-
- An adaptive feature is an inherited feature that helps an organism to survive and reproduce in its environment, i.e. increase fitness.

- **Adaptive features of hydrophytes:**

1 submerged leaves are divided ;
2 providing large area for, photosynthesis / absorption ;

3 leaves have large surface area ;
4 to float ;

5 (floating leaves so) little xylem/ little lignin ;
6 water provides support ;

7 little/ no roots /root hairs ;
8 roots for anchorage only / no need for roots to absorb water or mineral ions ;

9 little/ no, cuticle ;
10 no need to conserve water ;

11 stomata only on upper surface ;
12 only upper surface exposed to air/ to allow diffusion of gases ;

13 lots of air spaces (between cells) ;
14 for flotation/ buoyancy ;

15 floating leaves ;
16 to allow, diffusion/AW of (named) gas(es) ;

17 aerial roots ;
18 to allow roots to receive oxygen ;

- **Adaptive features of xerophytes:**

- few stomata / low stomatal density ;
- sunken stomata ;
- stomata close during the day and
- open at night ;
- rolled leaves ;
- thick epidermis / thick cuticle ;
- few / no / small, leaves ;
- hairs on leaves ;
- low rates of transpiration ;
- deep roots / AW ;
- to absorb water from the water table / AW
- long and spread out below the surface ;
- to absorb water when it rains ;
- root cells have low water potential ;
- to absorb water by osmosis ;
- from (very) salty soils / AW ;
- roots branch many times ;
- have many roots hairs ;
- to give a large surface area (for absorption of water) ;

- **Natural selection:**

- genetic variation within populations
- production of many offspring
- struggle for survival, there is competition for resources
- a greater chance of reproduction by individuals that are better adapted to the environment than others
- these individuals pass on their alleles to the next generation

- **Selective breeding:**

- selection by humans of individuals with desirable features
- crossing these individuals to produce the next generation
- selection of offspring showing the desirable features.

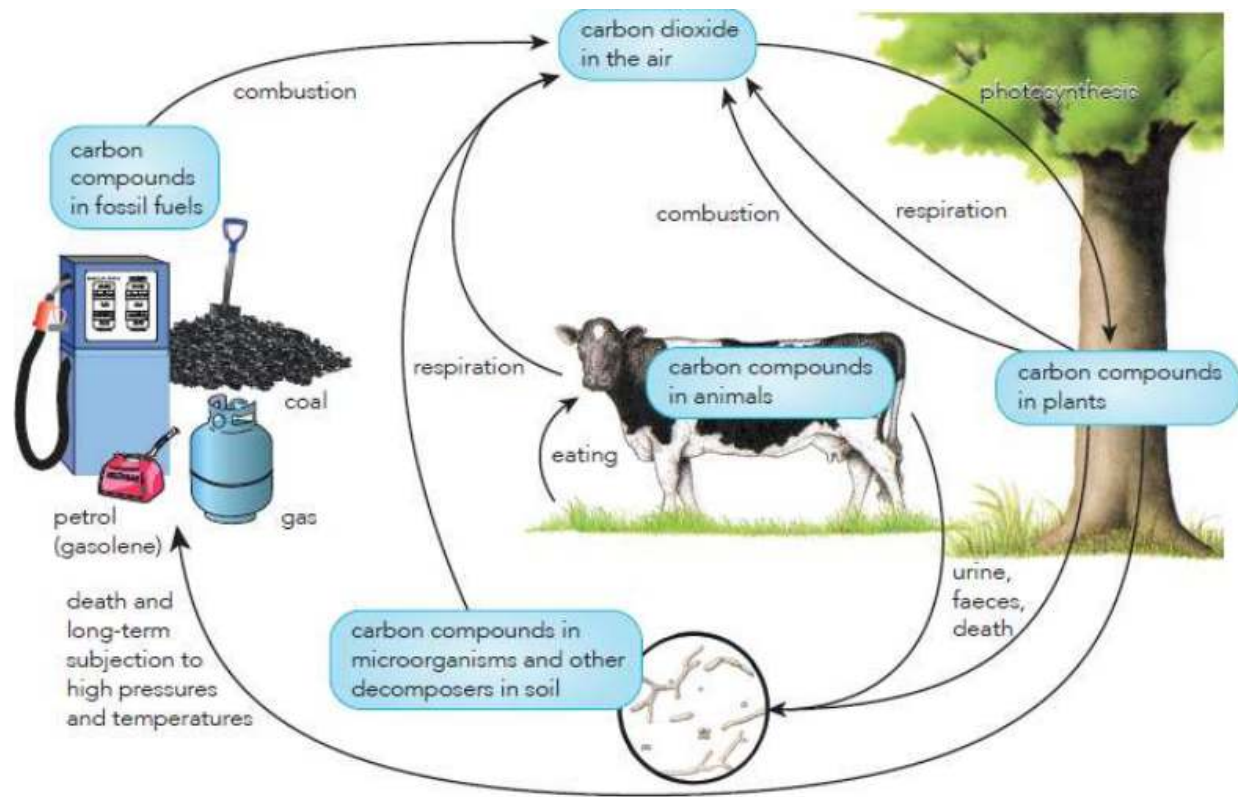
- Selective breeding by artificial selection is carried out over many generations to improve crop plants and domesticated animals.

- Adaptation is the process, resulting from natural selection, by which populations become more suited to their environment over many generations.
- The development of strains of antibiotic resistant bacteria is an example of natural selection.
- Difference between natural and artificial selection:
 - features are adaptive
 - caused by environment
 - slower changes
 - increase in fitness
 - explained: ability to survive AND reproduce (in natural environment)
 - maintain (genetic) variation/ less (genetic) variation in selective breeding
 - random mating
- Difference between artificial and natural selection:
 - human choice (rather than environmental pressures) / AW ;
 - less, diversity / variation ;
 - faster change ;
 - AVP ; e.g. mating is not random

CHAPTER 19

- The Sun is the principal source of energy input to biological systems.
 - The flow of energy through living organisms, including light energy from the Sun and chemical energy in organisms, is eventual transfer to the environment.
-
- A food web is a network of interconnected food chains.
 - A producer is an organism that makes its own organic nutrients, usually using energy from sunlight, through photosynthesis.
 - A consumer is an organism that gets its energy by feeding on other organisms.
 - Consumers may be classed as primary, secondary, tertiary and quaternary according to their position in a food chain.
 - A herbivore is an animal that gets its energy by eating plants.
 - A carnivore is an animal that gets its energy by eating other animals.

- A decomposer is an organism that gets its energy from dead or waste organic material.
- Advantages of using a pyramid of biomass rather than a pyramid of numbers to represent a food chain:
 - in a pyramid of numbers one large individual is shown in the same way as
 - one very tiny individual ;
 - biomass indicates how much food there is, available / left ;
 - biomass is an indicator of the energy available ;
 - pyramid of biomass is pyramid shaped whereas a pyramid of numbers is not always
- A trophic level is the position of an organism in a food chain, food web or ecological pyramid.
- Advantages of using a pyramid of energy rather than pyramids of numbers or biomass to represent a food chain:
 - it shows how much energy is available
 - it shows how much energy is passed through per meter square
 - other pyramids are not very informative, because we cannot be sure that one gram of biomass for one species contains the same quantity of energy as one gram of biomass of a different species.
- Transfer of energy from one trophic level to another is often not efficient because energy is lost between trophic levels. Energy is lost in respiration, movement, muscles contraction, heat. Energy is lost in faeces, urine. Not all animal or plant is digestible. Only 10% of the energy is transferred. As we move up the food chain lesser and lesser energy is available to support a population. Therefore a food chain is not longer than 5 trophic levels.



Carbon cycle

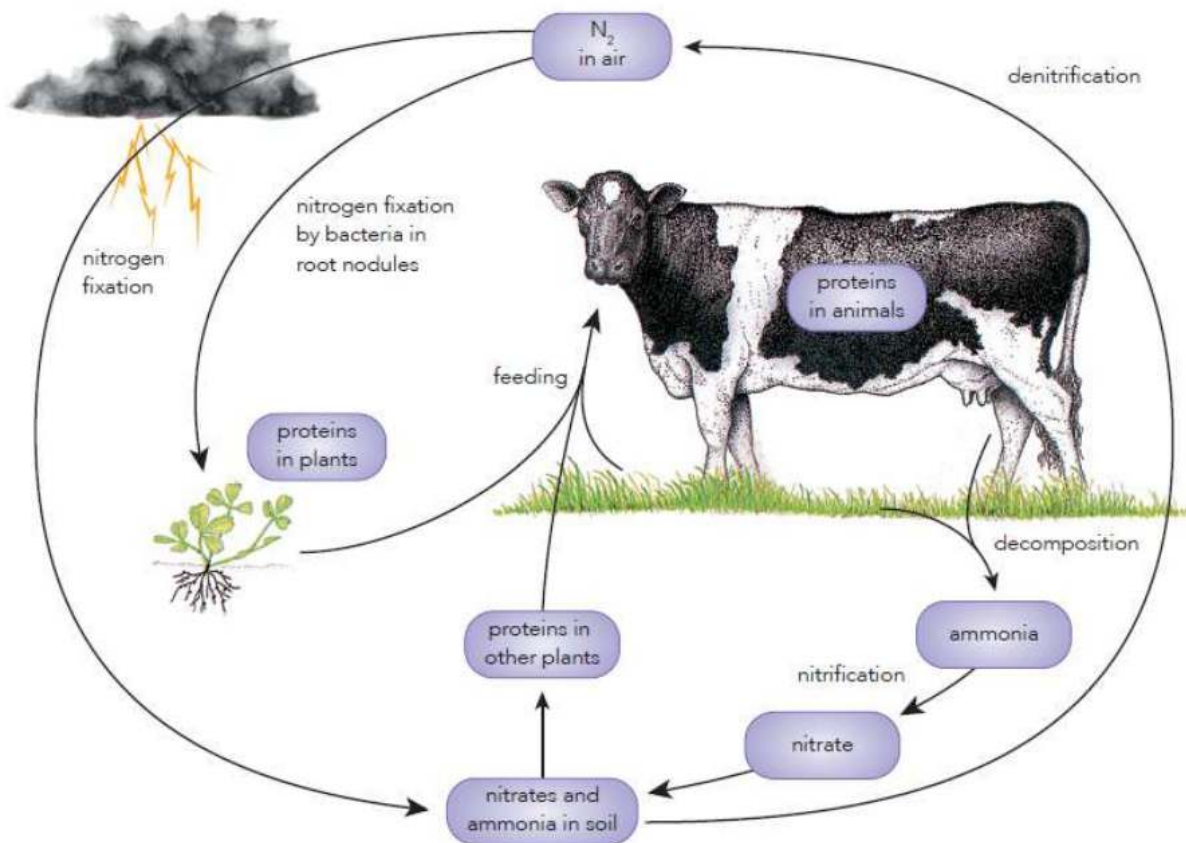


Figure 18.15: The nitrogen cycle.

- Nitrogen cycle:
 - nitrogen fixation occurs by lightning or by bacteria in root nodules, they convert nitrogen from air into ammonia.
 - ammonia is converted to nitrate ions in the process of nitrification, by nitrifying bacteria.
 - the plant absorb nitrate ions from the soil.
 - plant produce amino acids and protein from nitrate ions.
 - animals feed and digest protein.
 - deamination of protein occurs in animals and they excrete it as urea.
 - bacteria decompose plant and animal protein into ammonium ions.

- ammonium ions are converted to nitrogen in the process of denitrification, by denitrifying bacteria.
-
- A population is a group of organisms of one species, living in the same area, at the same time.
 - A community is all of the populations of different species in an ecosystem.
 - An ecosystem as a unit containing the community of organisms and their environment, interacting together.
 - Factors affecting the rate of population growth for a population of an organism:
 - food supply
 - competition
 - predation
 - disease
-

CHPATER 20

- How humans have increased food production:
 - agricultural machinery to use larger areas of land and improve efficiency
 - chemical fertilisers to improve yields
 - insecticides to improve quality and yield
 - herbicides to reduce competition with weeds
 - selective breeding to improve production by crop plants and livestock
- Disadvantages of monoculture:
 - loss of biodiversity
 - outbreaks/ spreading of diseases/ pests/ plagues
 - endangered/ extinction of species
 - disruption of food chain
- Advantages of monoculture:
 - higher revenue.
 - easier to manage.
 - yields maximization.
 - specialised production.
 - increased productivity and efficiency.

- loss of habitats/ habitat destruction
- deforestation
- disrupted soil fertility/ infertility of soil
- desertification
- soil erosion (by water/ wind)
- increased pollution
- competition for resources
- flooding
- disrupted nutrient cycling ;
- Disadvantages of intensive livestock production:
 - loss of biodiversity ;
 - deforestation / habitat loss ;
 - use of pesticides ;
 - release of, (named) greenhouse gases / carbon emissions ;
 - named consequence ; e.g. climate change / global warming
 - effect of waste (faeces and urine) on, waterways / crops ;
 - disease spreading to, wild populations / humans ;
 - use of antibiotics and (spread of) antibiotic resistance ;
 - providing food for livestock requires monocultures ;
 - idea that livestock production is an inefficient use of, crop plants / energy
- Advantages of intensive livestock production:
 - help to provide more food for people
 - sometime provide food more cheaply than extensive systems
 - fewer people go hungry

- take up less land
 - more land could be left in its natural state, providing habitats for other species
-
- Biodiversity is the number of different species that live in an area.
 - Reasons for habitat destruction:
 - increased area for housing, crop plant production and livestock production
 - extraction of natural resources
 - freshwater and marine pollution
 - Through altering food webs and food chains, humans can have a negative impact on habitats.
 - Effects of deforestation:
 - habitat destruction
 - reducing biodiversity
 - extinction
 - loss of soil
 - flooding
 - increase of carbon dioxide in the atmosphere
-
- Effect of pollution and excess fertilisers in aquatic ecosystems:
 - lake/ river pH decrease
 - aluminium ions become mobile
 - nutrients leached
 - shell damage
 - fish/ frog fail to reproduce
 - aquatic plants die
 - disruption of food chain
 - loss of biodiversity

- low pH toxic to aquatic animals
- fish produce mucus which blocks gills.
- Effect of plastic on aquatic ecosystem:
 - plastic remains /persists / lasts a long time/ not decomposed ;
 - swallowed/ ingested/eaten/ cannot be digested/ blocks gut ;
 - caught, around/ strangle/trapped/ entangled/ smother/ suffocate / injure/ cut / trap/ stuck in, organism AW ;
 - plastic blocks light for, photosynthesis ;
 - may, contain/release, (oil-soluble) toxins / poisons / harmful chemicals;
 - blocks the flow of water in streams or rivers ;
 - so less aeration of water/reduces concentration of (dissolved) oxygen ;
 - destruction of, habitat/ ecosystem/ food chain ;
 - idea of bioaccumulation/ biomagnification ;
 - trapped / stationary water acts as a breeding site for mosquitoes ;
 - AVP ; e.g. visual pollution /releases hormone-like chemicals / less oxygen from photosynthesis
- Effect of plastic on terrestrial ecosystem:
 - chokes / strangles / traps / blocks digestive systems / AW (of animals)
 - ingested ;
 - reference to, chemical exposure / fumes / toxins ;
 - (plastic) accumulates in an organism / is passed down a food chain ;
 - (described) habitat destruction ;
 - e.g. plastic covers the habitats (plastic) blocks (light / water for) photosynthesis (for land plants) ;
 - visual pollution ;
 - (plastic) block roots / prevents root growth ;
 - remain in the ecosystem (for a very long time) ;

- Methane and carbon dioxide are green house gases. They trap infrared radiation emitted from the sun towards the earth. This keeps the planet warm enough to support life. If level of these green house gases rise, more infrared radiation is trapped, this increase the average temperature of earth. Resulting in enhanced green house effect, leading to climate change.
- Eutrophication of water:
 - increased availability of nitrate and other ions
 - increased growth of producers
 - increased decomposition after death of producers
 - increased aerobic respiration by decomposers
 - reduction in dissolved oxygen
 - death of organisms requiring dissolved oxygen in water

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- Sustainable resource is one which is produced as rapidly as it is removed from the environment so that it does not run out.
 - Some resources can be conserved and managed sustainably, e.g. forests and fish stocks.

- Why an organism becomes endangered:

- climate change
- habitat
- destruction
- hunting
- overharvesting
- pollution
- introduced species

- **Reasons why a plant species can become endangered species:**

- plant hunting
- disease
- lack of pollinators
- competition from introduced species
- deforestation
- climate change
- desertification
- pollution

- increase in grazing
- habitat destruction
- removing too many of a plant species

- **Reasons why a fish species can become endangered species.**

- habitat destruction ;
- climate change ;
- (described) overfishing / hunting ;
- food chain disrupted (described);
- overconsumption (by humans) ;
- (named) pollution ;
- introduced diseases / species ;

- **Reasons why a animal species can become endangered species.**

- 1 habitat loss / deforestation ;
- 2 large mammals need large amounts of space ;
- 3 (named example of) climate change ; e.g. desertification

Use of land for

- 4 land developed for agriculture ;
- 5 land developed for, housing / transport / factories
- 6 land used for extraction of (named) material(s)
- 7 competing with humans for space ;

- 8 idea that populations are isolated ;
- 9 hunt / trophies / high value / poaching ;
- 10 laws are not enforced to protect from hunting

- 11 lack of education ;
- 12 poisoning / pollution ;
- 13 need a large quantity of food / lack of food sources / starvation ;
- 14 slow reproduction rate / hard to breed in captivity ;

- **How endangered species can be conserved:**

- monitoring and protecting species and habitats
- education

- captive breeding programmes
- seed banks
- **How endangered species can be conserved:**
 - monitoring / AW, population(s) / individual(s) ;
 - habitat, protection / restoration ;
 - reducing / prevention, of pollution ;
 - removal / AW, of alien species ;
 - preventing colonisation by alien species ;
 - hunting ban / prevent poaching ;
 - government / legislation, to protect species ;
 - create, exclusion zones / reserves (so not disturbed by people) ;
 - specific, times when / areas where, hunting / AW, not allowed
 - international agreements to limit trade ;
 - removal to, zoos / botanical gardens / wildlife parks ;
 - captive breeding / breeding programme (in situ or ex situ) ;
 - seed banks / frozen zoos / cryopreservation / AW ;
- **How can we prevent overfishing:**
 - 1 education / awareness ; Accept commercials / advertising / tax consumer
 - 2 reduced demand (to eat from unsustainable fish stocks) / public pressure / campaigning ;
 - 3 steps taken by fisherman voluntarily / AW ;
 - 4 (legal) quotas / treaties / licenses / laws / restricted catch weight ;
 - 5 ensuring sustainable population size / recovery of, endangered / specific, species
 - 6 **nursery zones (using this is much safer)/** no-catch zones / / protected areas / MPAs ;
 - 7 overflow of target species / increase in population outside zone / breeding recovery ;
 - 8 limited fishing season ;
 - 9 stock recovery / optimises breeding seasons ;
 - 10 fines;
 - 11 discourage / punish, poor practice
 - 12 restocking / captive breeding and release ;
 - 13 increases gene pool / number of young / reproductively-viable, fish ;
 - 14 fish farming ;
 - 15 alternative source of fish ;

- artificial insemination / IVF / use of surrogates / AW ;
 - reintroduction programmes ;
 - education / awareness ;
-
- Reasons for conserve animals
 - maintains natural habitat/ prevent, human interference/ development
 - prevention of extinction
 - less, hunting/ poaching/ killing
 - tourism/ economic reason
 - maintain (bio)diversity/ genetic diversity
 - maintain, gene, pool/diversity
 - maintain, food chain/ balanced ecosystems
 - available for scientific study
 - retain for future generations
 - maintain nutrient recycling
 - maintain, resource provision/ food/ drugs
 - prevent soil erosion/ flooding
 - education
 - Reasons to maintain ecosystems like wetlands
 1. to prevent extinction;
 2. maintain biodiversity;
 3. provide feeding grounds for animals / ref. to disruption of food, chains / web;
 4. provide, breeding grounds / places for breeding;
 5. provide, habitats / shelter;
 6. vulnerable to the effects of, development / drainage / AW;
 7. ref to flooding / natural disasters;
 8. ref to nitrogen cycle;
 9. ref to maintenance of water cycle;
 10. ref to carbon cycle; e.g. greenhouse gas / carbon storage / carbon sink
 11. waste disposal;
 12. provide, resources / food / fuel / drugs / raw materials;
 13. idea of areas for, recreation / (eco)tourism / education;
 14. ethical reasons / aesthetic reasons / AW;

- **Artificial insemination:**

1. collect semen / sperm (implied by context) ;
2. freeze semen / sperm, / use sperm bank / use sperm collected, far away/ long ago ;
3. selecting / washing / screening, of sperm / AW ;
4. use fertility drugs containing hormones for (super)ovulation ;
5. determine when female is about to ovulate ;
6. insert, semen / sperm, into female (vagina / uterus) ;

- **The risks to a plant species of having very small numbers:**

- high risk of extinction
- less reproduction
- high risk of genetic disease
- less variation
- small population so more vulnerable to diseases.
- reduced number of alleles
- less likely to adapt to change in environment

- **In vitro fertilisation (IVF):**

- collect / extract, eggs ;
- collect / extract, sperm ;
- use of donor, eggs / sperm ;
- fertilisation / described / IVF, outside the body / in a dish / in a laboratory ;
- embryo development in dish / AW
- screening of embryos ;
- inserting embryos into uterus ;
- give mother progesterone after inserting embryos (to maintain lining) ;

- **The disadvantage of captive breeding programme involving very small number of animals:**

- inbreeding
 - less genetic variation
 - reduced number of alleles
 - increased risk of genetic disease
 - can not reproduce
 - not enough animals to breed
 - less likely to adapt to change in environment
 - cost
-

CHAPTER 21

- Bacteria are useful in biotechnology and genetic modification due to their
 - rapid reproduction rate
 - ability to make complex molecules
 - few ethical concerns over their manipulation and growth
 - the presence of plasmids
- Anaerobic respiration in yeast is used to make ethanol for biofuel.
- Anaerobic respiration in yeast is used in bread making because it produces carbon dioxide and makes the dough/ bread rise.
- Pectinase digest the cell wall of the fruits. This release more juice from the fruit. It also helps to make the cloudy juice clear.
- Milk contains a sugar, lactose, which many people are intolerant to. Enzyme lactase is added to milk to breakdown lactose. This results in formation of lactose-free milk.
- Bacteria and fungi can reproduce exponentially inside fermenters, where they are provided with optimum temperature, pH, nutrients supply, and removal of waste product.
- Genetic modification is changing the genetic material of an organism by removing, changing or inserting individual genes.
- Process of genetic modification using bacteria:
 - isolation of the DNA making up a human gene using restriction enzymes, forming sticky ends
 - cutting of bacterial plasmid DNA with the same restriction enzymes, forming complementary sticky ends
 - insertion of human DNA into bacterial plasmid DNA using DNA ligase to form a recombinant plasmid
 - insertion of recombinant plasmids into bacteria
 - multiplication of bacteria containing recombinant plasmids inside a fermenter
 - expression in bacteria of the human gene to make the human protein

- Examples of genetic modification:
 - the insertion of human genes into bacteria to produce human proteins
 - the insertion of genes into crop plants to confer resistance to herbicides
 - the insertion of genes into crop plants to confer resistance to insect pests
 - the insertion of genes into crop plants to improve nutritional qualities

- **Advantages of genetically modifying crops.**
 - disease resistance ;
 - large(r) / fast(er), yield ;
 - drought resistance ;
 - ref to more desirable, product / increased income / AW
 - herbicide resistance ;
 - pest / insect, resistance ;
 - salt resistance ;
 - frost resistance ;
 - (named) nutritional enrichment ;
 - vaccine production ;
 - ref to benefits to, environment ;
 - ref to a qualified benefit to humans ; e.g. food shortage / described
 - health benefit
 - AVP ; growth modification e.g. short stems / adaptations to extreme environments / rapid improvement to crop / improvements using characteristic that are not present in natural population

- **Disadvantages of genetically modifying crops.**
 - reduced natural biodiversity ;
 - insect-resistant plants negatively affect pollinators ;
 - cross-pollinate into wild populations ;
 - (GM seeds are) expensive / AW ;
 - seeds need to be bought each season
 - ethical concerns of consumers ;
-

OTHERS

- Only birds lay eggs with hard shells **AND** have feathers (NOT wings because other animals also have wings).
- 1cm is 10000 micrometer.
- When air enters lungs it first passes through the larynx.
- Some hydrophytes have cuticle, but no hydrophytes have stomata on lower epidermis.
- Lightning converts atmospheric nitrogen to nitrate ions. Nitrifying bacteria converts soil nitrogen to nitrate ions.
- Penicillin is produced by fungus not bacteria.
- Penicillin is used to treat bacteria disease.
- Meiosis produce 4 haploid cells.
- Transpiration occurs in leaves **and stem**.
- Mucus in trachea is a chemical barrier.
- Rod cells have **greater sensitivity to light** and don't provide coloured vision.
- Cholera cannot spread by blood transfusion.
- Pollen grains **don't** travel down the pollen tube it is the male gamete/ nucleus that travel down the pollen tube.
- When calculating percentage change it is important write the negative sign when the change is decrease.
- Starch grains are stored in chloroplast.
- You should brush your teeth after a meal because the pH of mouth decreases and this acidic condition can dissolve the enamel.
- Gravity also affects the growth of a plant.
- During the mitosis process, the duplication of chromosomes occur before mitosis.
- Xylem is a tissue not organ.

 Biology Definitions